

RSA - public key cryptography

$$E(M, e) \leftarrow M_1 \xrightarrow{\text{client 1}} (e, n)$$

$$\square \xleftarrow{M_2} \text{client 2} (e, n)$$

$$\vdots$$

$$\square \xleftarrow{M_n} \text{client n} (e, n)$$

$$D(E(M_i, e), d) = M_i$$

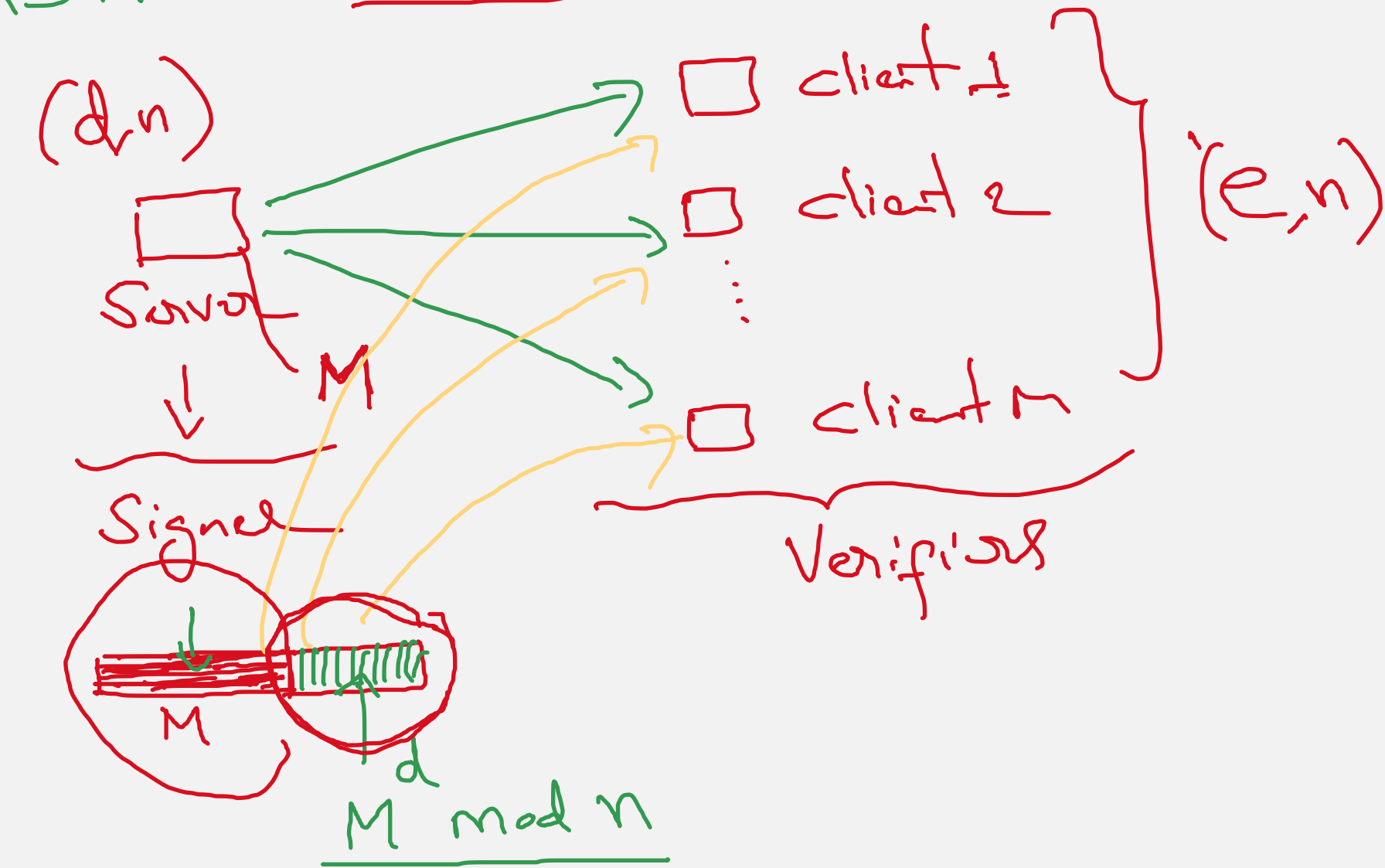
$$c = M^e \bmod n \quad \text{--- (1)}$$

$$M = c^d \bmod n \quad \text{--- (2)}$$

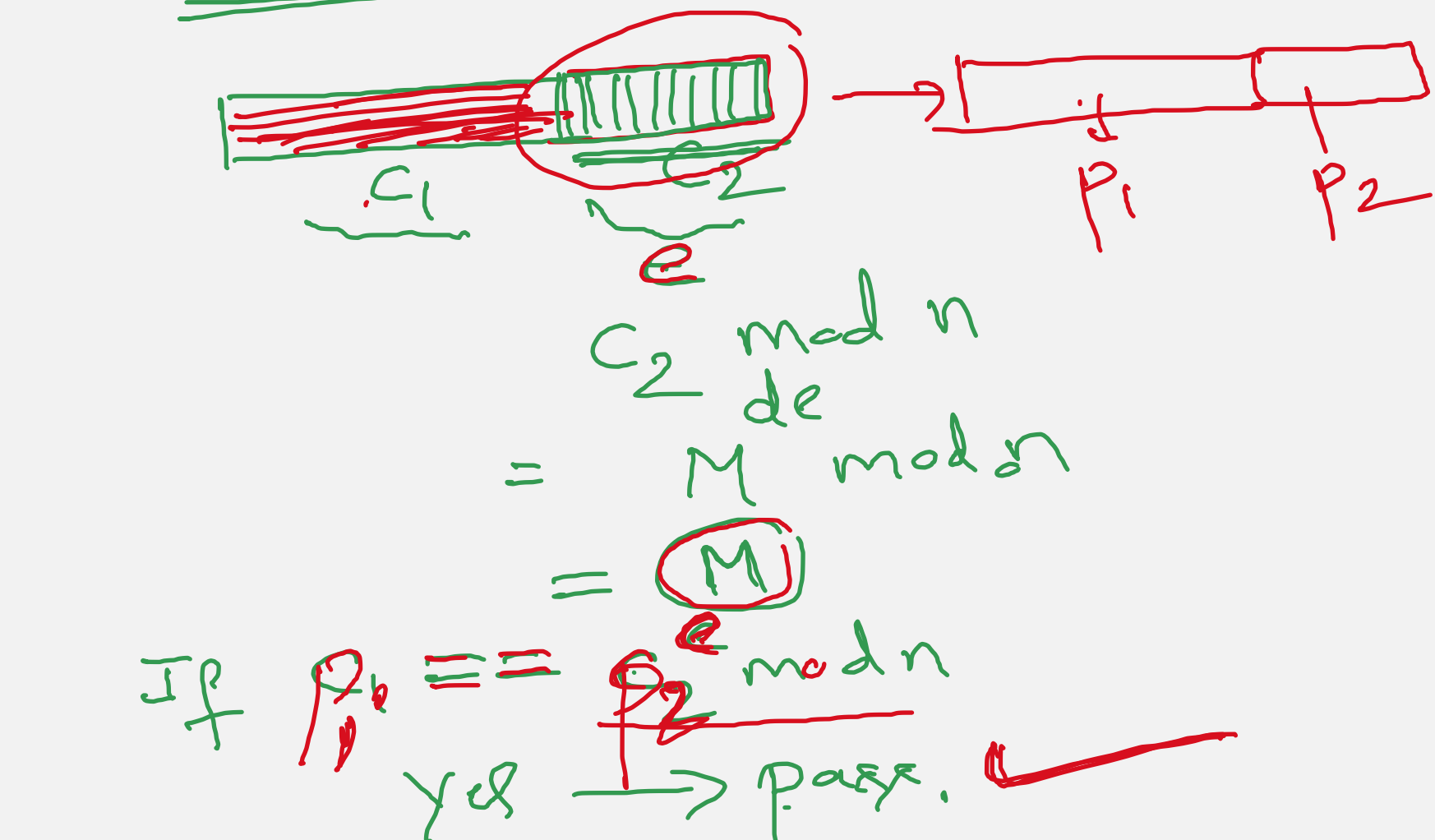
$$= M^{ed} \bmod n$$

$$= M$$

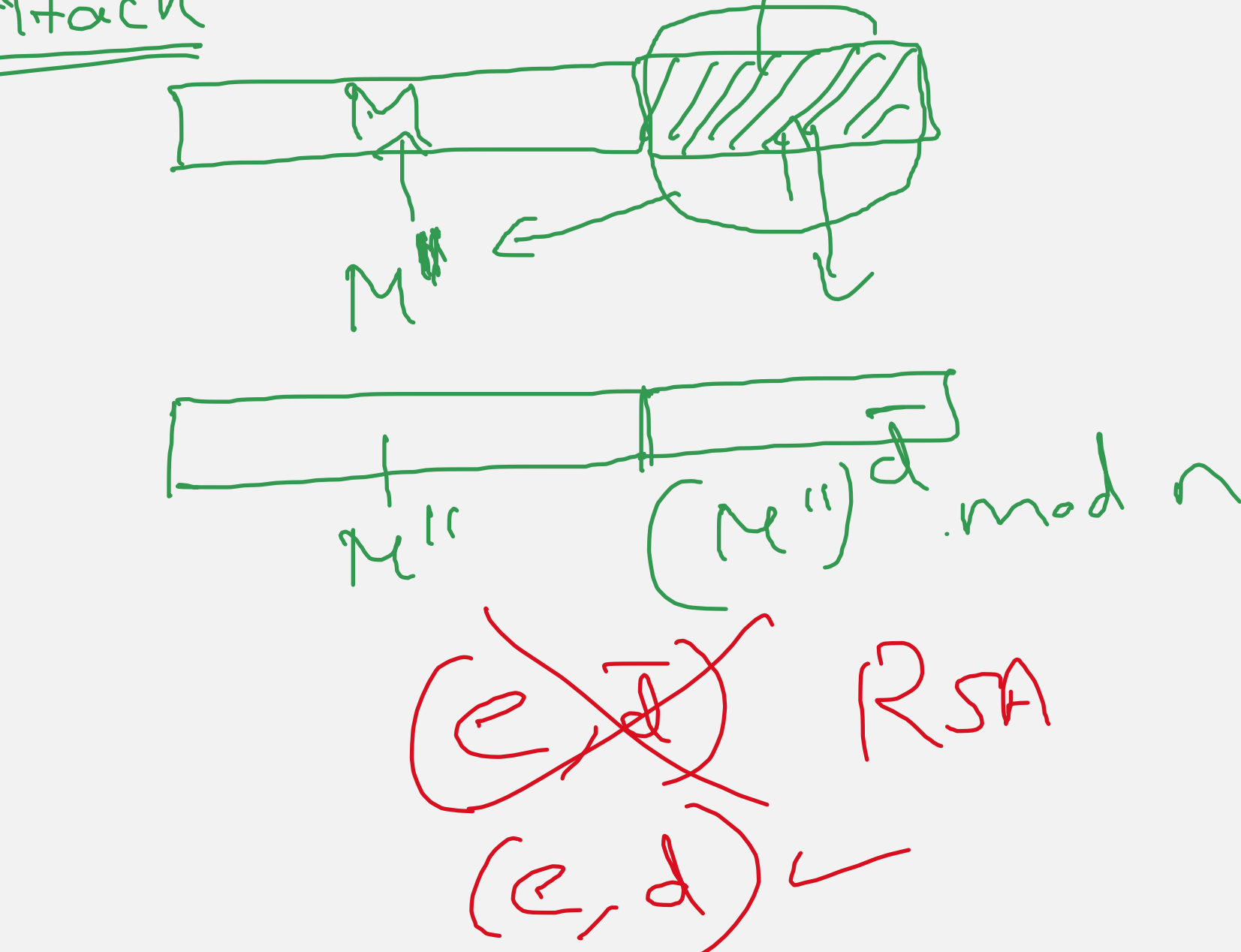
RSA -> Authentication



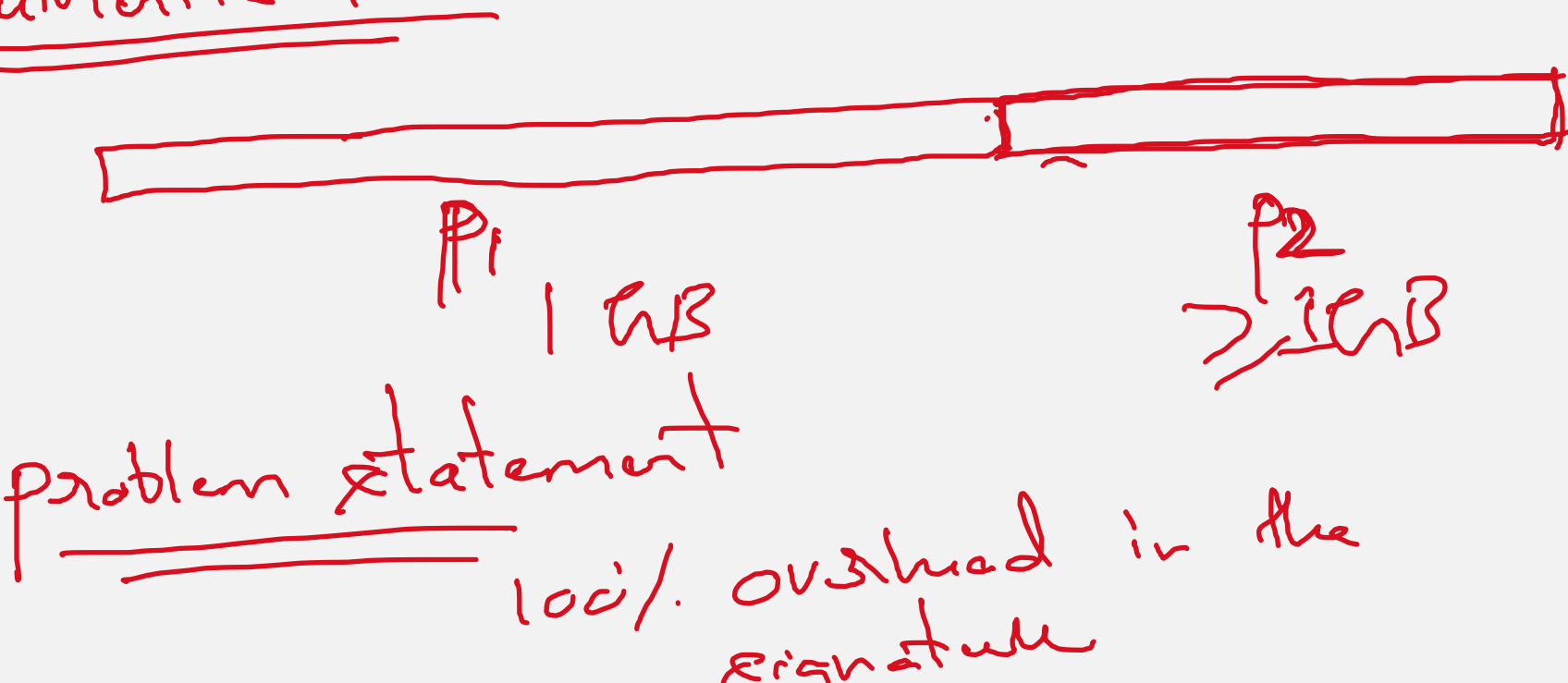
At client i



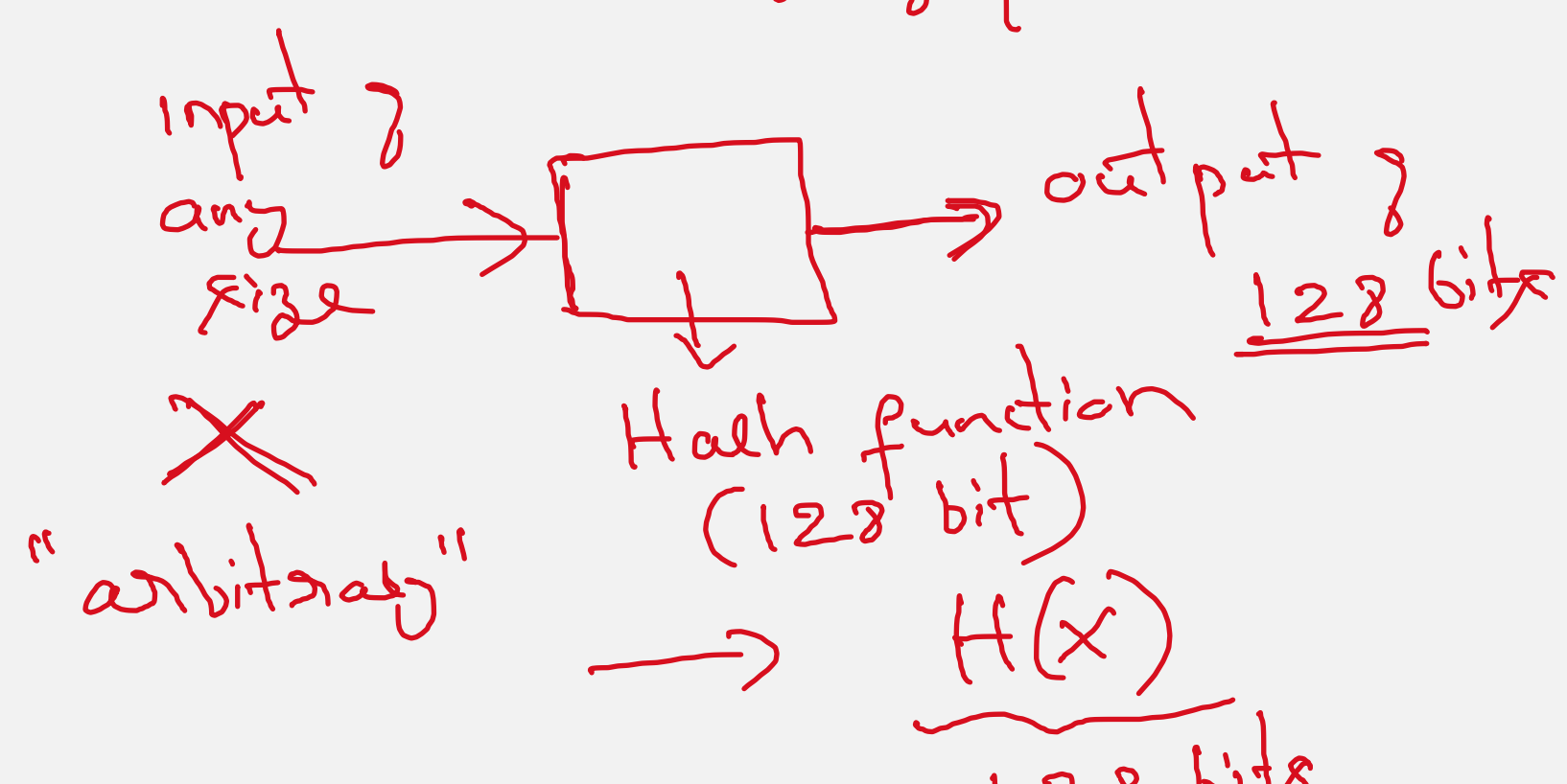
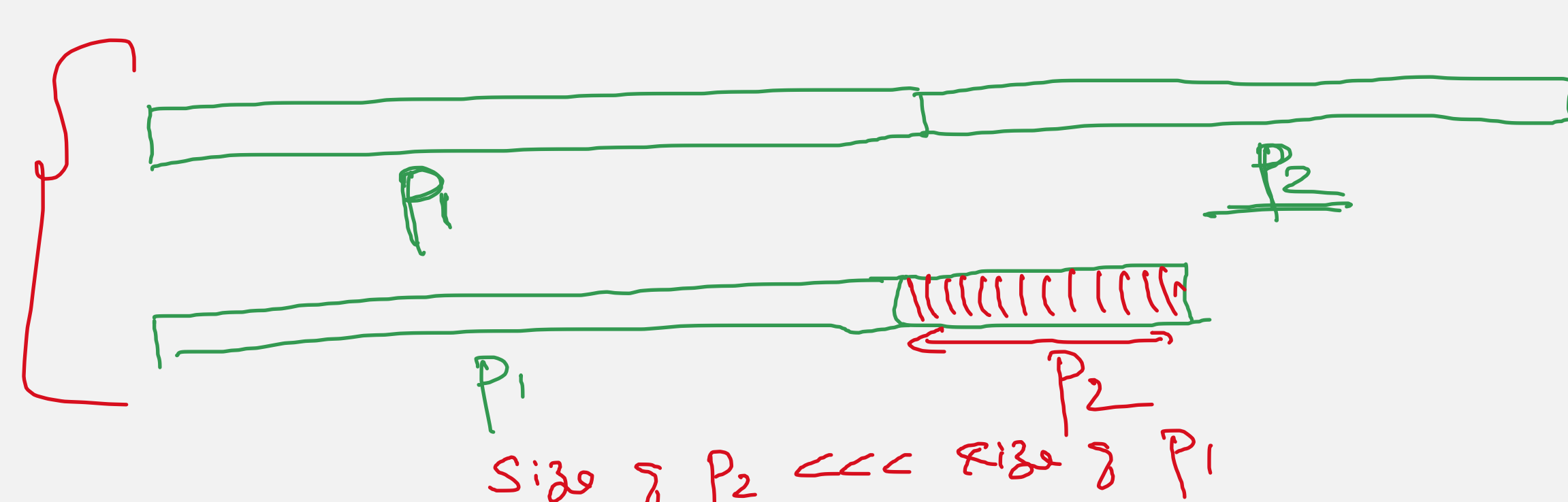
Attack



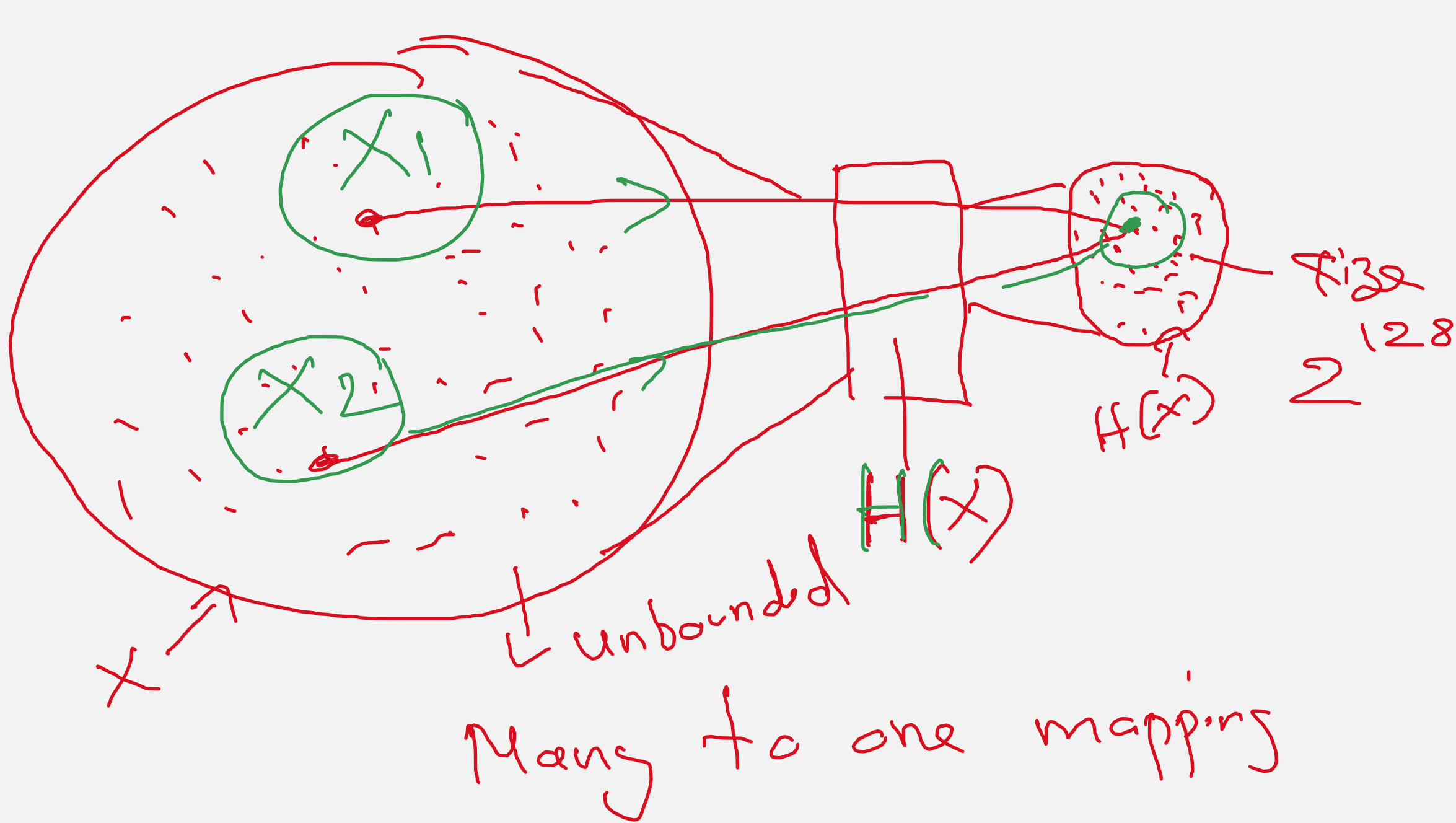
Authentication



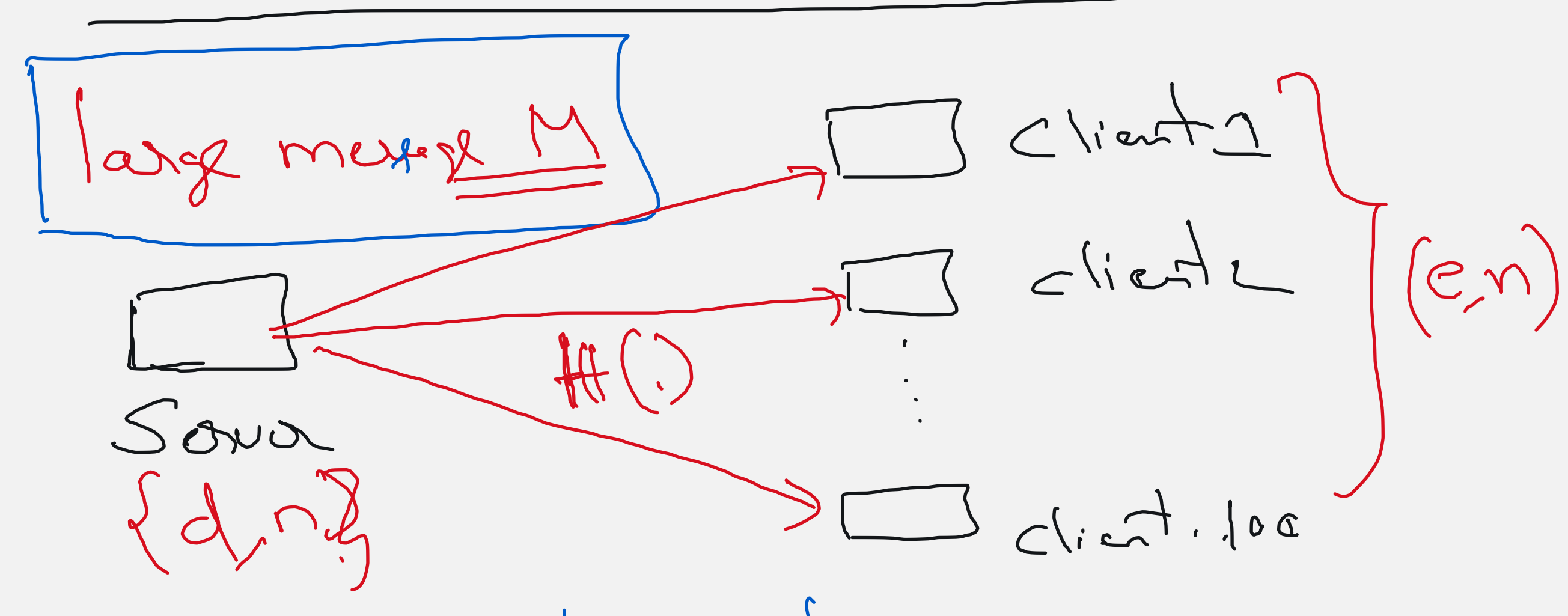
Hash functions



- 1) Invertible: Knowing $H(x)$ we cannot compute x backwards. (A)
- 2) If you know x_1 and $H(x_1)$ it is very difficult to find one more x_2 such that $H(x_2) = H(x_1)$. (B)
- 3) We cannot find two inputs x_1 and x_2 such that $H(x_1) = H(x_2)$. (C)

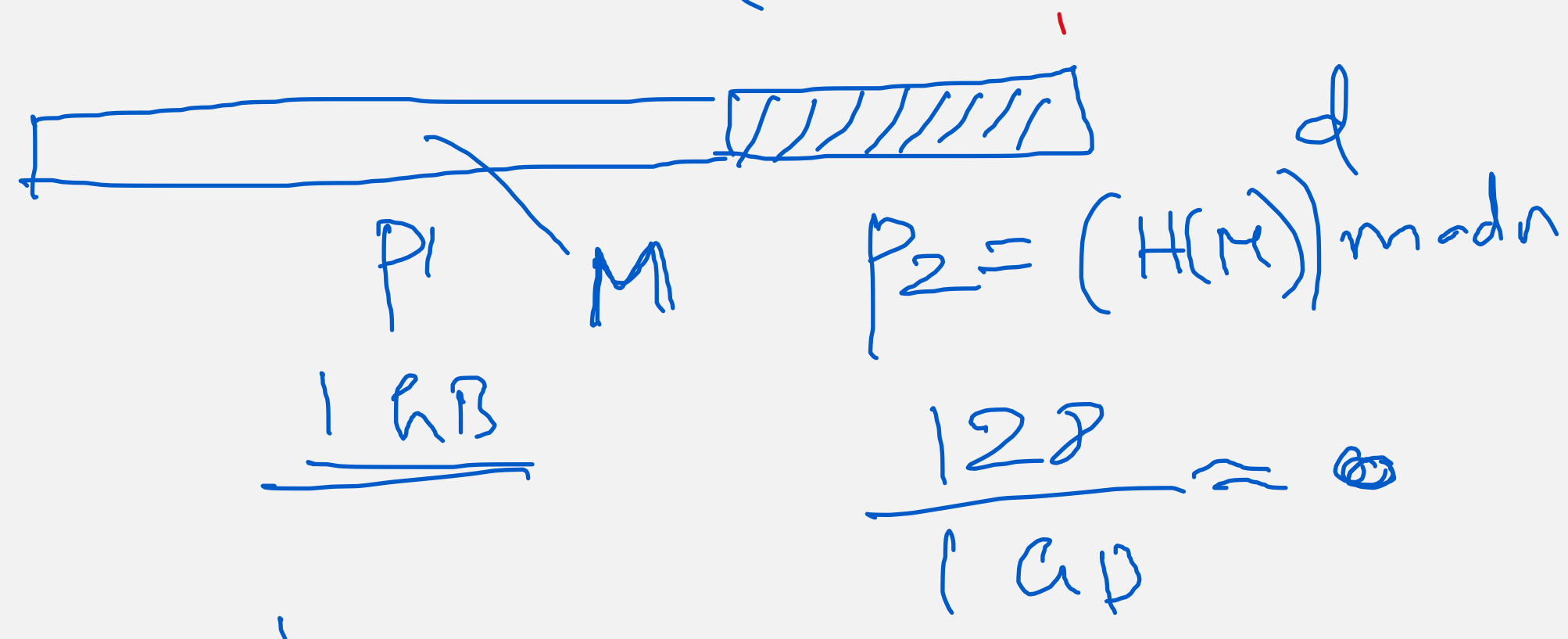


Authentication with Hash function

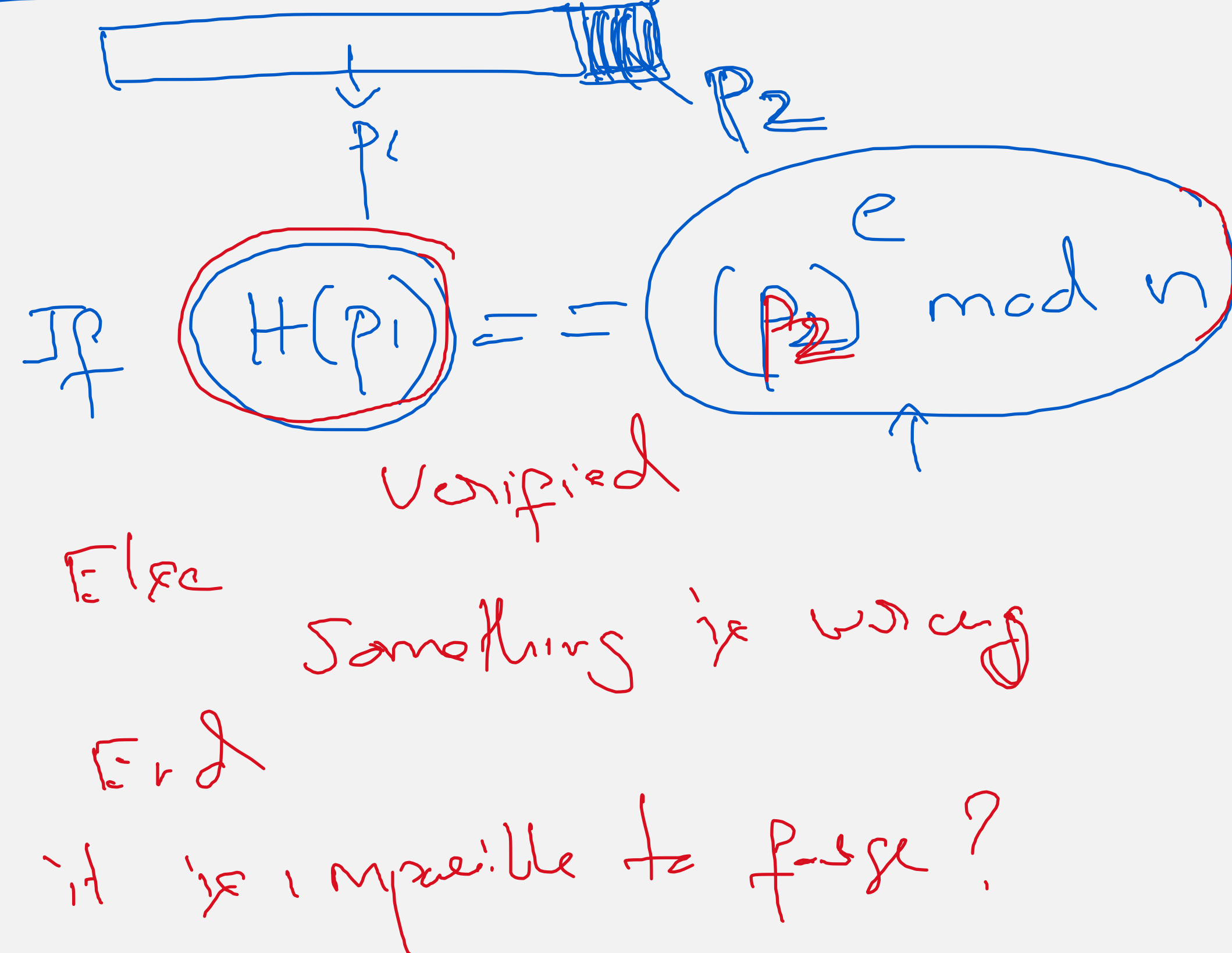


Signature creation step

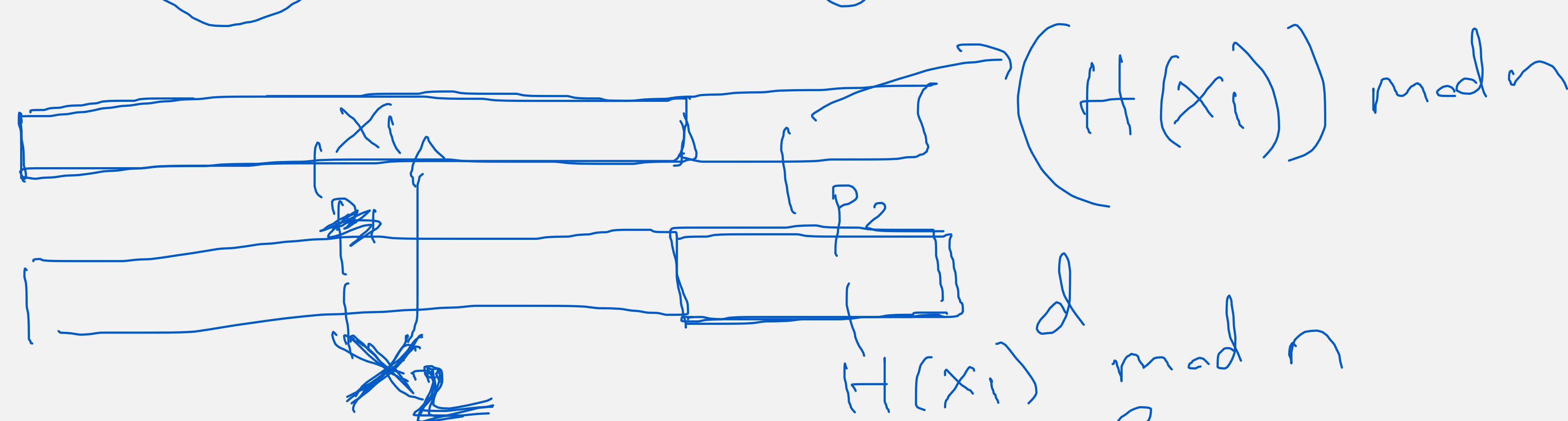
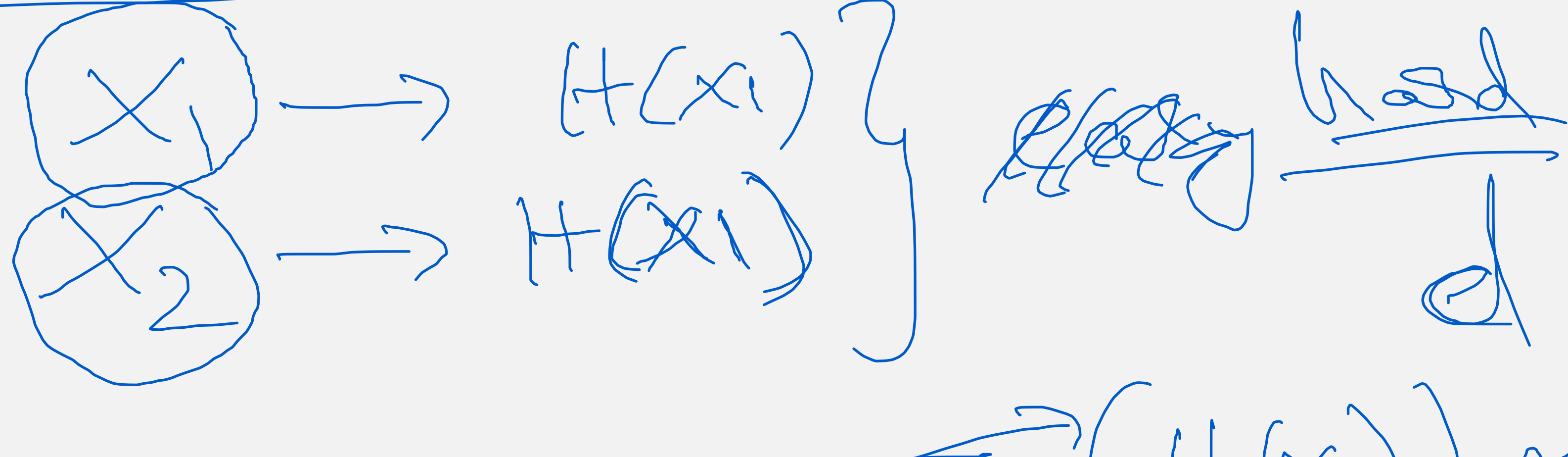
- 1) pick the message M
- 2) compute $H(M)$
- 3) compute $(H(M))^d \bmod n$



Verification



why it is impossible to forge?



if $H(x_2) == (H(x_1))^d \bmod n$ pass